EDITORS Gokhan Tur • Renato De Mori

SPOKEN LANGUAGE Understanding

Systems for Extracting Semantic Information from Speech

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SYSTEMS FOR EXTRACTING SEMANTIC INFORMATION FROM SPEECH

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In memory of Fred Jelinek (1932–2010)

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Dr. Tur is also the recipient of the Speech Communication Journal Best Paper awards by ISCA for 2004–2006 and by EURASIP for 2005–2006. Dr. Tur is the organizer of the HLT-NAACL 2007 Workshop on Spoken Dialog Technologies, and the HLT-NAACL 2004 and

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Foreword

Speech processing has been an active field of research and development for more than a half-century. While including technologies such as coding, recognition and synthesis, a long-term dream has been to create machines which are capable of interacting with humans by voice. This implies the capability of not merely recognizing what is said, but of understanding the meaning of spoken language. Many of us believe such a capability would fundamentally change the manner in which people use machines.

The subject of understanding and meaning has received much attention from philosophers over the centuries. When one person speaks with another, how can we know whether the intended message was understood? One approach is via a form of the Turing Test: evaluate whether the communication was correctly understood on the basis of whether the recipient responded in an expected and appropriate manner. For example, if one requested, from a cashier, change of a dollar in quarters, then one evaluates whether the message was understood by examining the returned coins. This has been distinguished as linguistic performance, i.e. the actual use of language in concrete actions.

This new book, compiled and edited by Tur and De Mori, describes and organizes the latest advances in spoken language understanding (SLU). They address SLU for human/machine interaction and for exploiting large databases of spoken human/human conversations.

While there are many textbooks on speech or natural language processing, there are no previous books devoted wholly to SLU. Methods have been described piece meal in other books and in many scientific publications, but never gathered together in one place with this singular focus. This book fills a significant gap, providing the community with a distillation of the wide variety of up-to-date methods and tasks involving SLU. A common theme throughout the book is to attack targeted SLU tasks rather than attempting to devise a universal solution to "understanding and meaning."

Pioneering research in spoken language understanding systems was intensively conducted in the U.S. during the 1970s by Woods and colleagues at BBN (Hear What I Mean-HWIM), Reddy and colleagues at CMU (Hearsay), and Walker and colleagues at SRI. Many of these efforts were sponsored by the DARPA Speech Understanding Research (SUR) program and have been described in a special issue of the IEEE Transactions on ASSP (1975). During the mid-1970s, SLU research was conducted in Japan by Nakatsu and Shikano at NTT Labs on a bullet-train information system, later switched to air travel information.

During the 1980s, SLU systems for tourist travel information were explored by Zue and colleagues at MIT and airline travel by Levinson and colleagues at AT&T Bell Labs and by Furui and colleagues at NTT Labs. The DARPA Air Travel Information System (ATIS) program and the European ESPRIT SUNDIAL project sponsored major efforts in SLU during the 1990s and have been described in a special issue of Speech Communication Journal (1994). Currently,

it is worth noting the European CLASSiC research program in spoken dialog systems and the LUNA program in spoken language understanding.

During recent decades, there has been a growth of deployed SLU systems. In the early stages, the systems involved recognition and understanding of single words and phrases, such as AT&T's Voice Response Call Processing (VRCP) and Tellme's directory assistance. Soon thereafter, deployed systems were able to handle constrained digit sequences such as credit cards and account numbers. Today, airline and train reservation systems understand short utterances including place names, dates, times. These deployments are more restrictive than research systems, where fairly complicated utterances were part of ATIS and subsequent systems.

During the early years of this century, building upon the research foundations for SLU and upon initial successful applications, systems were deployed which understood task-constrained spoken natural language, such as AT&T's How May I Help You? and BBN's Call Director.

The understanding in such systems is grounded in machine action. That is, the goal is to understand the user intent and extract named entities (e.g. phone numbers) accurately enough to perform their tasks. While a limited notion of understanding, it has proved highly useful and led to the many task-oriented research efforts described in this book.

Many textbooks have been written on related topics, such as speech recognition, statistical language modeling and natural language understanding. These each address some piece of the SLU puzzle. While it is impossible here to list them all, they include: *Statistical Methods for Speech Recognition* by Jelinek; *Speech and Language Processing* by Jurafsky and Martin; *Theory and Applications of Digital Speech Processing* by Rabiner and Schafer; *Fundamentals of Speech Recognition* by Rabiner and Juang; *Mathematical Models for Speech Technology* by Levinson; *Digital Speech Processing, Synthesis, and Recognition* by Furui; *Speech Processing Handbook* by Benesty *et al.*; *Spoken Language Processing* by Huang, Hon and Acero; *Corpus-based Methods in Language and Speech Processing* by Young and Bloothooft; *Spoken Dialogs with Computers* by De Mori.

The recent explosion of research and development in SLU has led the community to a wide range of tasks and methods not addressed in these traditional texts. Progress has accelerated because, as described by von Moltke: "No battle plan ever survives contact with the enemy." The editors state, "The book attempts to cover most popular tasks in SLU." They succeed admirably, making this a valuable information source.

The authors divide SLU tasks into two main categories. The first is for natural human/machine interaction. The second is for exploiting large volumes of human/human conversations.

In the area of human/machine interaction, they provide a history of methods to extract and represent the meaning of spoken language. The classic method of semantic frames is then described in detail. The notion of SLU as intent determination and utterance classification is then addressed, critical to many call-center applications. Voice search exploits speech to provide capabilities such as directory assistance and stock quotations. Question answering systems go a step beyond spoken document retrieval, with the goal of providing an actual answer to a question. That is, the machine response to "What is the capital of England?" is not merely a document containing the answer, but rather a response of "London is the capital of England."

There is an excellent discussion of how to deal with the data annotation bottleneck. While modern statistical methods prove more robust than rule-based approaches, they depend heavily on learning from data. Annotation proves to be a fundamental obstacle to scalability: application

to a wide range of tasks with changing environments. Active and semi-supervised learning methods are described, which make a significant dent in the scalability problem.

In addition to tasks involving human interaction with machines, technology has enabled us to capture large volumes of speech (in customer-care interactions, voice messaging, teleconference calls, etc.), leading to applications such as spoken document retrieval, segmentation and identification of topics within spoken conversations, identification of social roles of the participants, information extraction and summarization. Early efforts in speech mining were described in a special issue of the IEEE Transactions on Audio and Speech (2004).

Tur and De Mori have made a valuable contribution to the field, providing an up-to-date exposition of the emerging methods in SLU as we explore a growing set of applications in the lab and in the real world. They gather in a single source the new methods and wide variety of tasks being developed for spoken language understanding. While not yet a grand unified theory, it provides an important role in gathering the evolving state-of-the-art in one place.

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